Evaluating Data Centers as Local Energy System Assets: A System-Integrated PUE Approach

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Abstract

Data centers can export waste heat, provide flexible demand, and use storage to relieve local grid stress, but Power Usage Effectiveness (PUE) counts the electricity used for those services as overhead. System-Integrated PUE (siPUE) subtracts a single delay-adjusted, energy-equivalent credit from the PUE numerator. The credit is drawn from heat, flexibility, peak, and carbon-damage value in a jointly optimized design. It is zero when no services are active, so siPUE reduces exactly to PUE. The metric is evaluated with a two-stage design-and-operation model for four data-center archetypes under ten shared uncertainty states, with explicit accounting for heat, flexibility, land, and construction delay. The results show that every service-active design raises PUE in all 40 canonical scenarios, yet siPUE falls below the services-disabled comparator in 36 of 40 baseline scenarios and 23 of 40 scenarios under a shared min-max-regret design. Weighted by annual IT energy, a baseline service-active design raises average PUE from 1.17 to 1.24 while lowering siPUE to 1.14, and a worst-case-cost-regret design raises PUE to 1.34 while lowering siPUE to 1.15. The robust design earns 10.56 EUR/MWh IT of delay-adjusted system benefit, about 19% of the reference electricity price. The reference price sets the credit's magnitude but never its sign. Excluding carbon damage reverses the improvement, so both the size and sign of the credit are conditional on the design objective and the social-carbon valuation used. These findings are validated through internal property, held-out, and out-of-sample checks.

Keywords: Data centers, Power Usage Effectiveness, waste-heat reuse, optimization, local energy systems

Nomenclature

Metric terms			
PUE	Power Usage Effectiveness, $E_{\mathrm{FAC}}/E_{\mathrm{IT}}$	B_{sys}	Credited system benefit (EUR/yr)
siPUE	System-Integrated PUE	C_{ind}	Independent system-cost ledger (EUR/yr)
E_{FAC}	Facility electricity (MWh/yr)	E_{credit}	Delay-adjusted energy credit (MWh/yr)
E_{IT}	Delivered IT electricity (MWh/yr)	p_{ref}	Frozen reference price (EUR/MWh)
$C_{\mathrm{el}}, C_{\mathrm{trans}}$	Electricity-resource, network cost (EUR/yr)	a_s	Lifecycle availability factor, $a_s \in [0,1]$
C_{peak}	Peak-capacity cost (EUR/yr)	$[z]^+$	Positive part, $\max(0,z)$
$C_{\mathrm{CO2,dam}}$	Carbon-damage cost (EUR/yr)	$0, d$	Superscripts: comparator, design
$V_{\mathrm{heat}}, V_{\mathrm{flex}}$	Avoided heat, flexibility cost (EUR/yr)		
Design and optimization			
$x, \hat{x}$	Design variables, shared robust design	$C_s, C_s^\star$	Lifecycle market cost, scenario optimum
y_s	Hourly operational variables in state s	R_s	Normalized regret
$s \in \mathcal{S}$	Shared uncertainty state	Z	Worst-case regret
$K_{\mathrm{HX}}, K_{\mathrm{HP}}$	Heat-exchanger, heat-pump capacity		
Carbon accounting and regret normalization			
$\mathrm{Cap}_s^{\mathrm{applied}}$	Applied carbon cap (tCO2)	$\mathrm{Cap}_s^{\mathrm{policy}}$	Policy-target carbon cap (tCO2)
$E_{s,\min}^{\text{services-disabled}}$: comparator's minimum achievable emissions (tCO2)		ϵ	Regret-normalization floor
$E_{\mathrm{IT},s}^{\mathrm{PV}}$	PV-discounted lifetime IT energy (MWh)	k	Reference-price multiplier, siPUE(k)

ϕ	Flexible-load fraction	α^{base}	Base cooling-overhead ratio (pre-retrofit)
Appendix: network, heat, and storage parameters			
F_{12}, F_{23}	DC power flow, node pairs 1–2, 2–3 (MW)	F_{13}	DC power flow, node pair 1–3 (MW)
E^{co2}	Physical emissions (tCO2)	n	Network node index
t, T	Hour index; total representative hours	$\Delta t, w_t$	Timestep (h); cluster weight
$\varepsilon_n^{\text{th}}$	Node thermal emission factor	$G_{n,t}^{\text{th}}$	Thermal generation, node n, hour t (MW)
$\varepsilon^{\text{boil}}$	Boiler emission factor ($\text{tCO2/MWh}_{\text{th}}$)	D_t^{dh}	District-heating demand, hour t (MWh_{th})
H^{sold}	Delivered (sold) heat (MWh_{th})	K	District-heating price segments (count)
$\lambda^{\text{th}}, \eta^{\text{th}}$	Thermal-store self-discharge, round-trip efficiency	$\eta^{\text{ch}}, \eta^{\text{dis}}$	Battery charge, discharge efficiency
γ^{ups}	UPS reserve-floor fraction		

Abbreviations. CUE, Carbon Usage Effectiveness; DC-OPF, direct-current optimal power flow; DCRE, Data Centre Resource Effectiveness; DESS, Data centre Environmental Sustainability Score; DH, district heating; ERF/ERE, Energy Reuse Factor/Effectiveness; ETS, Emissions Trading System; HX, heat exchanger; HP, heat pump; REF, Renewable Energy Factor; WUE, Water Usage Effectiveness.

1. Introduction

Electricity demand from European data centers has become a material planning variable for transmission and distribution networks. Commission estimates put EU consumption at 76.8 TWh in 2018, with projections of 98.5 TWh by 2030 under a trend case and up to 160 TWh under a high-growth case [1]; more recent assessments suggest roughly 96 TWh in 2024 and nearly 168 TWh by 2030 [2]. Annual demand, operational ramps, peaks, heat rejection, cooling needs, and flexibility headroom are reshaped by traffic expansion and compute-intensive AI workloads [3, 4].

At this scale, data-center build-out becomes a local infrastructure and investment problem. Connection rights, congestion, heat offtake, flexibility-market eligibility, land-use approvals, permitting timelines, and community

acceptance increasingly determine which projects proceed [5, 6]. The EU energy transition adds a second pressure. Electricity and heat systems need flexibility, digital coordination, and sector coupling [7, 8], while sustainability reporting for large data centers is being formalized through Directive (EU) 2023/1791 and sector initiatives [9, 10]. Data centers can therefore act either as inflexible loads or as service-active assets that export waste heat, provide grid support, and manage dynamic AI-driven loads [11, 12].

The reporting problem is that PUE evaluates only internal electricity overhead. Heat pumps, heat exchangers, controllability, storage, and telemetry can increase facility electricity even when they reduce local system cost. This creates the KPI gap studied here. PUE observes the facility-energy burden, while the independent system-cost benefit remains outside the metric. siPUE is introduced as an indicator, expressed in PUE's own energy units, that translates that benefit into the same numerator language as PUE while preserving PUE's internal-efficiency role.

1.1. Related Work

The relevant literature spans data-center metrics, heat recovery, flexibility, integrated energy systems, and robust decision-making. PUE remains the dominant facility-efficiency ratio [13, 14], but critiques show that it omits flexibility, source effects, climate dependence, and wider sustainability dimensions [15, 16]. Parallel indicators such as CUE (Carbon Usage Effectiveness), WUE (Water Usage Effectiveness), ERF/ERE (Energy Reuse Factor/Effectiveness), and REF (Renewable Energy Factor) add specific dimensions through the ISO 30134 family and EU reporting practice [17, 18], but they do not provide a market-valued PUE numerator credit.

Waste-heat and district-heating studies establish data-center heat as a real system service [11, 19], while demand-response and flexibility studies show that IT load, cooling, and batteries can provide grid services under suitable market rules [20, 21]. Integrated energy-system papers quantify these services inside optimization models and show potential cost or emissions reductions [22, 23]. These studies motivate system valuation, but they leave open the standalone reporting question. How should an operator report a service-active design when PUE rises because the service itself uses electricity?

Broader scoring approaches, including DESS and DCRE, widen the resource and context boundary [24, 25], and recent heat and flexibility reviews reinforce the need for quantifiable energy-system indicators [26, 20]. The

combined, delay-adjusted heat, flexibility, peak, and carbon-damage value of a jointly optimized design remains outside a PUE-normalized reporting signal.

An established robust-optimization structure is used below as the uncertainty testbed. Minimizing worst-case regret follows the min-max-regret tradition [27, 28]. The methodological claim is therefore limited. PUE and siPUE responses are observed for the same normalized min-max-regret design choice.

1.2. Synthesis and Gap Identification

Table 2 positions the proposed metric against the closest reporting metrics and integrated-system modeling literature. The contribution lies in the accounting translation from local-system value into a PUE numerator credit.

Across this literature, the remaining gap is an accounting one. Current reporting metrics either measure one dimension or widen the resource boundary without translating local-system services into PUE's own energy units. Heat reuse, flexibility, and carbon impacts are also seldom valued with empirical electricity, heat, reserve, or ETS prices. None of the reviewed approaches feeds the combined, delay-adjusted value of a jointly optimized design's heat, flexibility, peak, and carbon-damage effects back into the PUE numerator.

1.3. Conceptual Framework and Research Questions

Energy-system context, data-center design decisions, and local energy-system outcomes are linked in the analysis. Policy documents, market data, and industry practice supply the institutional context summarized in Section 2; the model in Section 3 carries the structured, testable claims. The resulting model-facing research questions are:

1. RQ1: Can an indicator, expressed in PUE's own energy units, reveal the combined local-system value that a jointly optimized design creates through heat, flexibility, peak, and carbon damage, adjusted for construction delay, while reducing exactly to PUE when no services are active?
2. RQ2: How do shared min-max-regret design decisions shift traditional efficiency metrics and system-integrated indicators?

The min-max-regret design is treated as an uncertainty testbed, with investment-rule validation left outside the scope.

Table 2: Comparison of siPUE with adjacent data-center metrics and modeling approaches. WUE (Water Usage Effectiveness) and REF (Renewable Energy Factor) are omitted from the table: neither values an energy-system service, so both differ from siPUE in the same way that CUE does.

Approach	Primary function	Values local-system services?	PUE numerator credit?	Gap relative to siPUE
PUE [13, 18]	Facility electricity divided by IT electricity	No	Baseline ratio	Treats heat-pump, storage, and control electricity as overhead even when they create external value.
ERF/ERE [18]	Reused-energy reporting and energy-reuse effectiveness	Heat only, in energy terms	Partial reuse adjustment	Omits flexibility, peak, network, carbon-damage, and delay effects.
CUE [15]	Carbon emissions per IT energy	Carbon footprint only	No	Parallel intensity indicator; no numerator credit.
DESS [24]	Broader sustainability score	Partial, contextual	No	Qualitative and contextual; no numerator credit.
DCRE [25]	Resource-effectiveness view of energy, water, water stress, and climate context	Indirect	No	Broader resource boundary; no numerator credit.
Integrated energy-system optimization papers [22, 23]	System-cost or emissions optimization	Yes, inside the model	No	Quantifies system impacts while leaving standalone reporting-compatible numerator credits undefined.
siPUE (this study)	Numerator credit for service-active data centers	Yes: heat, flexibility, peak, and carbon damage, adjusted for construction delay	Yes	Converts the independent system-cost benefit into an energy-equivalent numerator credit and reduces to PUE when no services are active.

1.4. Contributions

siPUE is introduced as an indicator that converts local-system benefit into a PUE numerator credit. Its boundary, normalization, and reference-price rules define the conversion from monetized system value to energy-equivalent credit. The metric is tested in a two-stage model that combines scenario-perfect benchmarks with a shared normalized min-max-regret design and explicit ledgers for lifecycle delay, carbon, land, heat recovery, flexibility, storage, and DC-OPF network constraints. The evaluation spans four archetypes and ten shared uncertainty states, decomposes the credit by ledger component, and reports sensitivity to price, technology, land, and delay assumptions.

2. Data Centers in Local Energy Systems — Policy and Market Context

Data-center integration depends on macro policy, meso infrastructure, and micro finance. A service-active project becomes viable only when these levels align, as summarized in Figure 1.

At the macro level, the recast Energy Efficiency Directive imposes reporting obligations on large data centers and codifies PUE, Renewable Energy Factor, and Energy Reuse Factor [9]. The accompanying delegated regulation sets the reporting template [18], and the Climate Neutral Data Centre Pact adds voluntary renewable-supply and heat-reuse targets [10]. The Green Deal and "Fit for 55" package raise the value of efficiency, waste-heat reuse, and flexibility [30, 7]. The action plan on digitalizing the energy system treats flexibility, interoperability, and data availability as prerequisites for renewable integration [8]. Carbon pricing works through the same channel. By raising the marginal cost of fossil generation, EU ETS prices increase the relative value of demand reduction, heat reuse, and electrified heat [7]. Because these KPIs now enter due diligence, ESG scoring, and financing conditions [18], a metric boundary that omits system services distorts investment signals.

At the meso level, grid capacity is usually the binding constraint. Large steady loads can trigger reinforcement, congestion charges, import caps, or paused connection queues [5, 6]. Projects that credibly reduce peak stress are therefore better placed in permitting and connection negotiations. Land and heat-network access constrain projects in parallel, since integration equipment expands the footprint and heat value depends on network proximity,

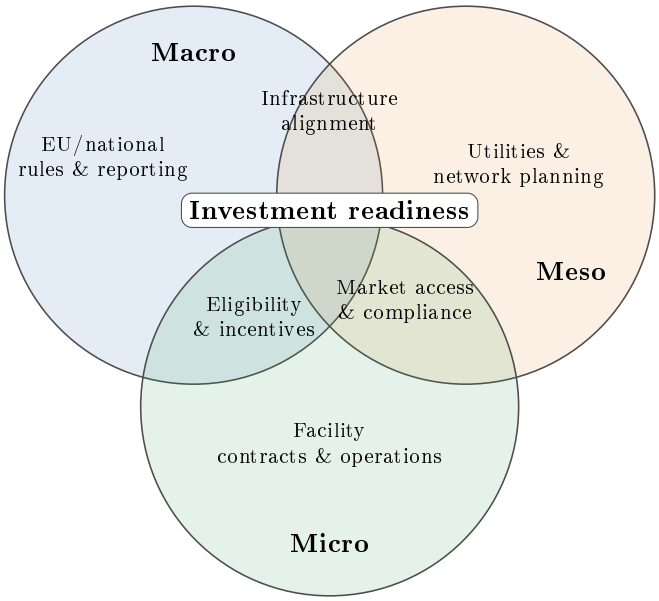


Figure 1: Conceptual macro–meso–micro governance structure. Investment readiness emerges when regulatory signals, infrastructure feasibility, and project-level execution align. This structuring is adapted from the multi-level perspective in socio-technical transitions literature [29] to fit the data-center energy-integration context.

temperature requirements, pipe capacity, and tariff treatment [11, 31]. Timing is the dominant meso risk. Data centers can be built faster than grid reinforcements or district-heating extensions, so service revenue may arrive after capital is committed. Renewable-resource geography reinforces this pattern. Nordic and energy-hub settings improve overhead and heat-cascading economics, while dense urban clusters offer heat demand but tighter grid and land constraints [32, 33].

At the micro level, heat exchangers, heat pumps, storage, controllability, and telemetry raise capital cost but diversify revenue. Contracted heat or flexibility revenue supports cheaper debt; speculative service revenue pushes financing toward equity, which encourages staged integration in which connection rights or heat offtake are secured first and further storage or controllability deferred [31, 34]. Because PUE and related KPIs enter the same due-diligence process, projects that add integration electricity can look weaker precisely when they create system value [13, 16].

Local-system value is available only where a service can be contracted and measured. Heat recovery draws on 25–40°C cooling streams and uses heat exchangers or heat pumps to reach district-heating or process-heat temperatures; commercial value depends on offtake contracts, tariffs, and responsibility for temperature lift [11, 35, 31]. Electrical flexibility comes from controllable IT and cooling load and from batteries installed for reliability, with value set by product definitions, telemetry, settlement rules, and grid-access or congestion-risk effects [12, 20, 21]. Renewable procurement and energy-hub co-location can anchor siting decisions, but their value remains place-specific because land, grid access, and heat-network proximity determine where integration can be realized [32, 33, 36].

Integration can stall when reporting, infrastructure planning, and finance give different signals. A project can satisfy decarbonization policy while still facing grid-connection limits, spatial constraints, or uncertain service cash flow. PUE remains the dominant reporting metric [17, 13], but it classifies all non-IT electricity as overhead. Heat pumps, storage, and controllable cooling therefore worsen PUE even when they create external value [16, 15]. The siPUE metric preserves the PUE numerator and denominator logic and adds a separate credit for monetized local-system benefits, leaving private revenues and costs outside the credit. The model follows the same boundary. Demand follows IT workload profiles, and delay enters through lifecycle availability. Electricity-side constraints use hourly balance and a three-node DC-OPF. Heat, flexibility, storage, carbon, land, and lease costs enter through explicit

physical and accounting ledgers.

3. Method: siPUE definition and two-stage model

siPUE and the optimization framework used to test it are defined in this section. The PUE structure is retained, while the independent system-cost benefit is translated into an energy-equivalent numerator credit. The benefit includes avoided resource, transmission, peak, and carbon-damage cost; private heat and flexibility revenue remain in the market-cost objective.

The model evaluates capital decisions (heat exchangers, heat pumps, flexibility capacity, batteries, land, and the locked 15% efficiency retrofit, a discrete cooling-overhead-reduction choice defined in Appendix A.1 with its own capex and opex cost) together with hourly operation. Lifecycle market cost includes electricity, carbon-market payments, capex, land, congestion, service revenues, nodal import constraints, and construction-delay effects.

The main reference case is the *services-disabled* comparator. In this case, all services are disabled, while the 15% efficiency retrofit and the physical constraints remain unchanged. A separate *identity* solve has no services and no efficiency investment. It checks that siPUE reduces to PUE when no services are active and sets the design-independent reference price p_{ref} used for unit conversion (Section 3.2). The *baseline* case represents a service-active design, while the *robust* case uses one shared min–max-regret design for each archetype. Table 5 defines all five solve modes. The baseline, robust, and scenario-perfect cases use the same scenario-specific carbon cap as the services-disabled comparator:

$$\text{Cap}_s^{\text{applied}} = \max\left(\text{Cap}_s^{\text{policy}},\ E_{s,\min}^{\text{services-disabled}}\right), \tag{1}$$

where $\text{Cap}_s^{\text{policy}}$ targets a 5% reduction in attributable emissions relative to the services-disabled comparator's own unconstrained, cost-minimizing emissions in scenario s, evaluated at that comparator's own chosen efficiency level. The second argument, $E_{s,\min}^{\text{services-disabled}}$, is the same comparator's minimum achievable emissions, obtained by minimizing emissions directly as the objective.

The max operator guards against scenarios in which the 5% target is physically unreachable for that design. In practice, across the canonical rows, the applied cap exceeds the policy target by at most 0.10% (mean 0.05%), so the enforced cap remains within a tenth of a percentage point of the stated

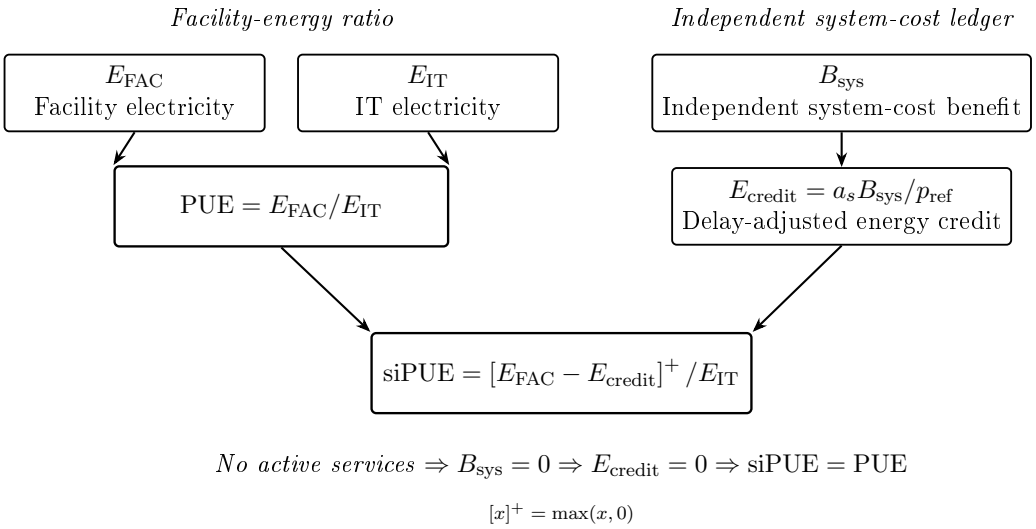


Figure 2: Metric logic for siPUE. PUE is the internal facility-energy ratio computed from measured electricity only. siPUE subtracts a delay-adjusted, energy-equivalent credit associated with the independent system-cost benefit from the facility-energy term before normalizing by IT electricity. The positive-part operator ensures that the adjusted numerator cannot become negative. When no services are active, $E_{\mathrm{credit}} = 0$ and siPUE = PUE.

5% reduction in every row. The cap-relaxed diagnostic in Section 4.2 tests whether this constraint drives the carbon-damage credit.

3.1. Definition and interpretation of siPUE

siPUE treats the data center as both a consumer and a prosumer of the local energy system. It reports both roles in one energy-equivalent number; Figure 2 summarizes the resulting metric logic. PUE captures the consumer role, the electricity drawn to serve IT load. The credit term captures the prosumer role, the heat, flexibility, peak relief, and avoided carbon damage supplied back to the system. This second role is priced through the independent system-cost ledger defined below (Table 3), so the prosumer credit reflects value delivered to the system. Private cash flow from the same prosumer activity (heat and flexibility revenue, capex, land, financing) is tracked separately, in the lifecycle market-cost objective (Table 3).

PUE is defined as

$$\mathrm{PUE} = \frac{E_{\mathrm{FAC}}}{E_{\mathrm{IT}}}, \tag{2}$$

where E_{FAC} is modeled facility electricity and E_{IT} is delivered IT electricity.

siPUE keeps this denominator and adjusts the numerator by a system-benefit credit. Let superscript 0 denote the services-disabled comparator and d a service-active design. The independent system-cost ledger is

$$C^d_{\text{ind}} = C^d_{\text{el}} + C^d_{\text{trans}} + C^d_{\text{peak}} + C^d_{\text{CO2,dam}} - V^d_{\text{heat}} - V^d_{\text{flex}}, \tag{3}$$

where C_{el}, C_{trans}, C_{peak}, and $C_{\text{CO2,dam}}$ are electricity-resource, network, peak-capacity, and carbon-damage costs, and V_{heat} and V_{flex} are avoided heat and flexibility costs. V_{heat} is avoided boiler fuel cost plus avoided variable O&M only; it excludes any carbon or emissions term. V_{flex} is a fixed avoided reserve-capacity value, independent of carbon pricing. The carbon-damage value of the emissions avoided by displacing boiler heat or thermal generation is credited separately through $C_{\text{CO2,dam}}$. This term uses the same physical emissions ledger as V_{heat}'s displaced fuel, but applies its own cost basis: social carbon cost times avoided tCO2. The credited system benefit is

$$B^d_{\text{sys}} = C^0_{\text{ind}} - C^d_{\text{ind}}. \tag{4}$$

Positive terms in B_{sys} are avoided independent system costs; negative terms are additional system burdens. Components are aggregated before any clipping, so a design with negative net system benefit receives a negative credit and siPUE rises relative to PUE.

Heat and flexibility revenues remain in the market objective, while carbon damage enters only the independent system ledger. ETS payments and carbon damage use the same emissions data but different prices, so emissions are not counted twice.

All quantities entering Eqs. 3–4 are annual and undiscounted: E_{FAC} and E_{IT} in MWh per representative modeled operating year, C_{ind} and B_{sys} in EUR per year for that same year.

The delay-adjusted energy credit is

$$E_{\text{credit}} = a_s \frac{B_{\text{sys}}}{p_{\text{ref}}}, \tag{5}$$

where p_{ref} is a fixed unit-conversion price in EUR/MWh and $a_s \in [0, 1]$ is the dimensionless lifecycle availability factor after construction delay. In the canonical run, $p_{\text{ref}} = 56.63$ EUR/MWh, the electricity-resource marginal cost at the data-center node in the identity solve. The identity solve carries zero efficiency investment and zero service capacity, so its marginal price cannot

Table 3: Accounting boundary for the siPUE system-benefit term (upper panel) and treatment of the three carbon mechanisms (lower panel).

Included in B_{sys}	Excluded from B_{sys}
Electricity resource-cost changes, transmission or network-cost changes, independent peak-capacity effects, carbon-damage changes, avoided heat-system cost, and avoided flexibility-system cost.	Private heat and flexibility revenues, IT service revenue, ETS or carbon-market payments, capex, land lease or opportunity cost, private demand charges, taxes, subsidies, and financing terms.

Carbon mechanism	Purpose	In optimization?	In siPUE credit?
ETS payment	Private compliance and dispatch cost in the market ledger.	Yes	No
Carbon-damage value	External damage value in the independent accounting ledger.	Ledger only	Yes
Physical carbon cap	Row-level planning constraint shared by the services-disabled comparator, baseline, robust, and scenario-perfect modes.	Yes	No direct monetary credit

be shifted by the design choices siPUE evaluates. Anchoring p_{ref} to the services-disabled comparator would let that comparator's locked efficiency retrofit and carbon-cap treatment influence a conversion factor common to every case. ETS payments and carbon damage are excluded from p_{ref} because they are handled in separate ledgers.

B_{sys} is annual, while a_s accounts for construction delay over the asset life. Post-construction siPUE sets $a_s = 1$ and is used for annual reporting. Delay-adjusted siPUE retains a_s and is used for investment decision making. siPUE is

$$\text{siPUE} = \frac{[E_{\text{FAC}} - E_{\text{credit}}]^+}{E_{\text{IT}}}, \tag{6}$$

with $[z]^+ = \max(0, z)$. Internal property checks verify nonnegativity, reduction to PUE when no services are active, delay behavior, carbon-ledger separation, scenario count, scale normalization, benefit monotonicity, and reference-price monotonicity.

The positive-part operator in Eq. 6 is a guard that binds if the credited system value exceeds total facility electricity. Reported siPUE values stay away from zero. The smallest value is 0.96 under the low-conversion-price sensitivity (Table 4), so the guard remains slack in the canonical results and siPUE remains a smooth function of B_{sys}. A value below one remains possible when the conversion price is low enough (Section 3.2). In ledger terms, credited system value can exceed facility electricity in energy-equivalent terms. The site is then a net contributor on the independent ledger while still consuming more electricity than its IT load. Because siPUE is a ledger indicator, the reference price must be reported alongside the physical PUE value.

siPUE also rewards value efficiency in integration. A design with twice the installed service capacity of another can score worse if the additional capacity delivers less system value per unit of facility electricity consumed. This behavior is intentional: additional equipment improves the score only when its system value is large enough relative to its own electricity burden. Section 4 shows this property in the reported designs: the robust design earns roughly twice the absolute system benefit of the baseline design and still scores marginally worse.

Table 4: Reference-price sensitivity for the canonical robust rows. Dispatch and ledgers are held fixed; only the value-to-energy conversion factor changes. Average siPUE uses IT-energy weights across archetypes.

Reference-price case	Interpretation	Average siPUE
$0.5p_{\text{ref}}$	Low conversion price; stronger energy credit	0.96
$1.0p_{\text{ref}}$	Canonical frozen reference price	1.15
$1.5p_{\text{ref}}$	High conversion price; weaker energy credit	1.21

3.2. Reference-price sensitivity

The reference price is the main conversion choice in siPUE. Holding design and operation fixed, a multiplier k gives

$$\text{siPUE}(k) = \frac{[E_{\text{FAC}} - a_s B_{\text{sys}}/(k p_{\text{ref}})]^+}{E_{\text{IT}}}. \tag{7}$$

p_{ref} is held constant across designs and uncertainty states. Table 4 changes only Eq. 7; dispatch, heat delivery, emissions, PUE, and B_{sys} are unchanged.

The sensitivity shows that p_{ref} scales the magnitude of the credit even when every physical and monetary ledger is unchanged. With $p_{\text{ref}} > 0$ and $a_s \geq 0$, the sign of E_{credit} follows the sign of B_{sys} for any conversion price. Changing p_{ref} therefore preserves whether siPUE lies above or below PUE and scales only the size of the gap. The conversion price is a disclosed reporting convention. siPUE should be reported with the reference-price rule and a sensitivity band.

3.3. Reporting protocol and computability

siPUE requires more than facility meter readings. While PUE requires only facility and IT electricity measurements, siPUE also requires a standardized no-services reference case, avoided-cost values for the surrounding energy system, and a carbon-damage value. These inputs sit outside the facility fence, which places siPUE in a different reporting class from PUE, ERF, CUE, or REF.

The operator can measure the physical quantities: facility and IT electricity, delivered heat H^{sold}, contracted and delivered reserve, peak import, and installed service capacity. The system side must be supplied by the network or heat-system operator, or set by the regulator as published defaults. These

values include avoided heat-system and reserve-capacity unit values, peak-capacity and network unit costs, the reference price p_{ref}, and the social cost of carbon. siPUE is therefore a jointly verified indicator, computed from operator-metered physical quantities and externally published unit values, and it should be reported with those unit values attached.

The no-services reference case is fixed by convention: service equipment is set to zero while the efficiency retrofit and other physical parameters remain at their as-built values. A facility with no service capacity is therefore its own comparator and reports siPUE = PUE.

3.4. Two-stage design model

Let $s \in \mathcal{S}$ index the shared uncertainty states, x denote design variables, and y_s denote hourly operational variables. Stage 1 solves a scenario-perfect benchmark:

$$C_s^\star = \min_{x,y_s} C_s(x, y_s), \tag{8}$$

where C_s is lifecycle market cost. Stage 2 chooses one shared design $\hat{x}$ per archetype using min-max regret [27, 28]. The ten states are unweighted, and each archetype is solved independently. Regret is normalized by each scenario's present-value-discounted lifetime IT energy $E_{\mathrm{IT},s}^{\mathrm{PV}}$:

$$R_s(\hat{x}) = \frac{C_s(\hat{x}, y_s) - C_s^\star}{\max(\epsilon, E_{\mathrm{IT},s}^{\mathrm{PV}})}. \tag{9}$$

The robust design minimizes $Z = \max_{s\in\mathcal{S}} R_s(\hat{x})$ directly for each archetype. Table 5 lists the five solve modes and states which of them are reported as design cases.

The canonical run sets the efficiency reduction to 0.15 in the services-disabled comparator, baseline, robust, and scenario-perfect cases. Optimized capacities are heat exchanger, heat pump, flexibility, battery energy and power, and associated land use. The identity solve keeps service capacity and efficiency reduction at zero. Re-solving the fair services-disabled comparator in every one of the 40 canonical rows with efficiency instead chosen freely from the same discrete candidate set (0% or 15%) selects 0.15 in all 40 rows: the comparator's own unconstrained economics already favor the retrofit, independently of any service, so the locked value matches what free optimization would choose anyway.

Table 5: Solve modes used in the study and the role of each. "Reported as a design case" identifies the modes whose PUE and siPUE values appear in the results.

Solve mode	Definition	Reported as a design case?	Quantity it supplies
Identity	No services and no efficiency investment.	No	Checks that siPUE = PUE and sets the reference price p_{ref}.
Services-disabled comparator	No services; the 15% efficiency retrofit and all physical constraints are retained.	Yes (reference case)	Provides the reference ledger C^0_{ind} in Eq. 4 and the carbon-cap basis in Eq. 1.
Baseline	Assumed baseline service-active portfolio (Table 11).	Yes	Provides baseline PUE, siPUE, and system benefit.
Robust	One shared min–max-regret design per archetype across the ten states.	Yes	Provides robust PUE, siPUE, system benefit, and credit decomposition.
Scenario-perfect	Best design for each individual scenario.	No	Provides the benchmark $C^\star_s$ in Eq. 9. KPIs are omitted because this benchmark produces one design per scenario.

3.5. Operational constraints and ledgers

Hourly operation is a linear program. The facility balance links grid import, battery operation, IT load, cooling, and heat-pump electricity. Heat export is bounded by heat-exchanger and heat-pump capacity, recoverable heat, pipe limits, and district-heating acceptance. Flexibility is modeled as reserve availability, with routine activation outside the settlement basis [20, 21]. Deliverability is enforced through IT-load headroom, controllable capacity, unused battery inverter capacity, stored energy above the UPS reserve, and parasitic overhead.

The electricity system is a stylized three-node DC-OPF with thermal generation, renewable availability, line capacities, and nodal settlement prices (Figure 3). Carbon costs follow the market/independent ledger split in Table 3. Land is represented as lease or opportunity cost tied to core facility, cooling, heat, flexibility, and battery footprints. The heat side is a heat-recovery chain with storage and price-segmented sale (Figure 4).

3.6. Data and scenario design

Figure 5 summarizes the workflow from inputs to reported metrics. The model uses a calibrated 8,760-hour, three-node time series compressed into seven representative days. Stratified medoid clustering selects representative days from seasonal strata using standardized fuel-price, load, renewable, net-load, calendar, ambient-temperature, and district-heating features. Cluster

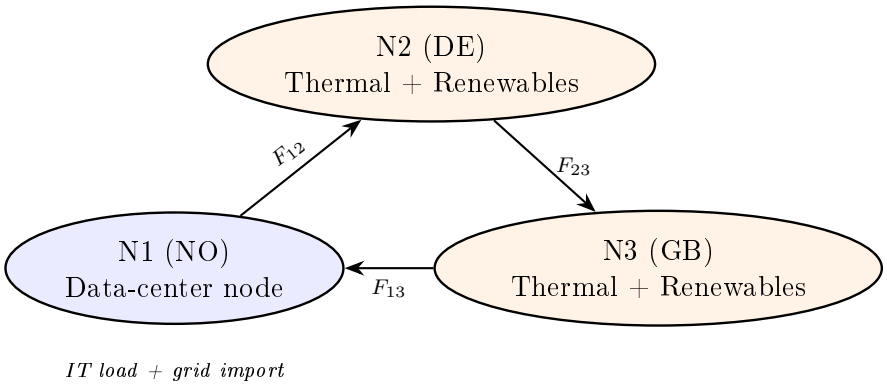


Figure 3: Stylized three-node DC-OPF electricity network. Each node represents one modeled country group, N1 (Nordics: Norway, Sweden, Denmark, Finland, Iceland), N2 (Central Europe: Germany, Belgium, Netherlands, France), and N3 (United Kingdom); one representative country is shown per node. F_{12}, F_{23}, and F_{13} are the linearized DC power flows between node pairs; arrow direction is illustrative, since flows are signed variables that can reverse hour to hour. The data-center node shown here (N1) is illustrative. Each archetype's own scenario location determines which node it actually attaches to. Thermal generation capacity is 5,500 MW at each node. Calibrated base line capacities, before the per-scenario congestion multiplier (Table 8), are 938.3 MW (F_{12}), 422.6 MW (F_{23}), and 1345.3 MW (F_{13}); see Table A.15 (Appendix A.10) for the full network parameter list.

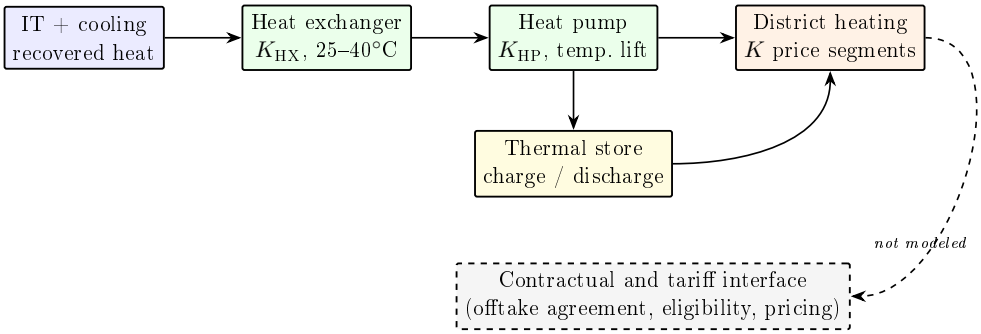


Figure 4: Heat-recovery chain as modeled, with the commercial interface that determines whether the chain is investable. Recovered heat from IT and cooling loads passes through the heat exchanger and heat pump to become delivered heat, is optionally buffered in a thermal store, and is sold into up to K district-heating price segments. The solid boxes are the physical and price-segment representation inside the optimization model. The dashed box is the contractual and tariff interface (Section 2): a design can clear every technical step and still fail commercially if the offtake contract is missing, but this interface is exogenous to the model and enters only through the district-heating price and acceptance capacity.

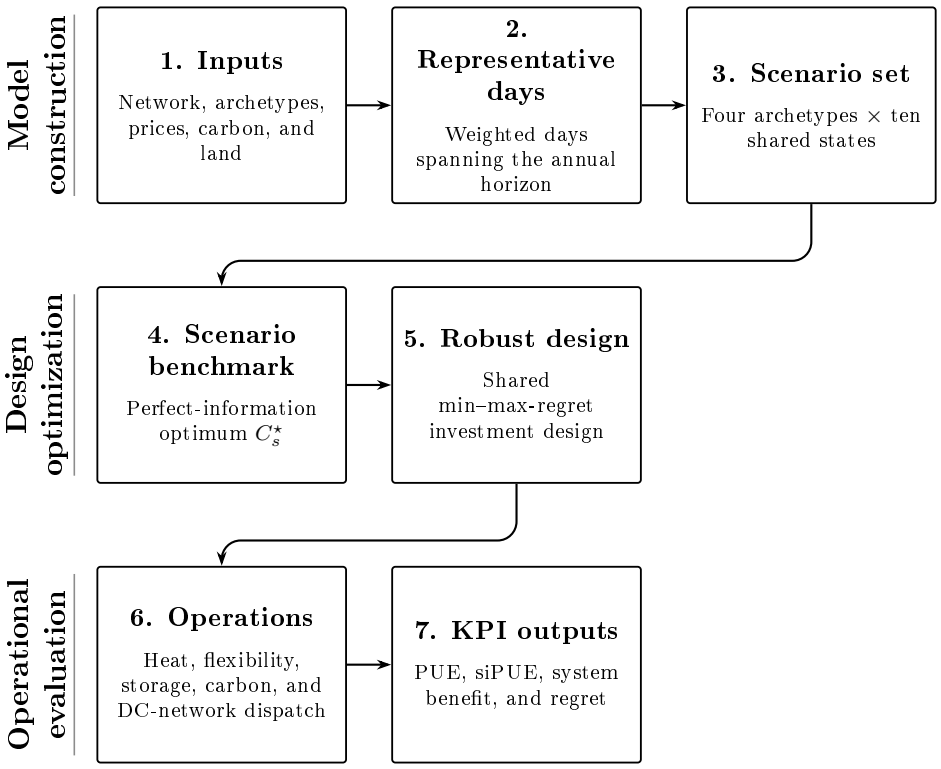


Figure 5: Model workflow from input construction to performance evaluation. Each phase reads left to right; the elbow arrows wrap from the end of one phase to the start of the next. Inputs are reduced to weighted representative days and combined with the shared scenario set. Scenario-specific perfect-information benchmarks $C_s^\star$ are used to formulate the shared min–max-regret design, which is then evaluated using the operational model and reported through energy, system-benefit, and regret metrics.

weights preserve annual hours. Fixed annual terms enter once outside the representative-day weighting. The seven days are concatenated into one operating horizon. Battery state of charge and peak import are evaluated across the combined sample, with state carried across day boundaries.

The canonical scenario set has four archetypes (AI/HPC, hyperscale cloud, colocation, and enterprise/edge), each facing ten shared uncertainty states, giving 40 rows, and each row is solved in every mode of Table 5. States vary service prices, heat conditions, construction delay, carbon caps, and operating conditions. Table 6 defines the four archetypes. Archetype base recoverable-heat fractions are 0.20–0.50 [11, 37]; state multipliers raise effective LP inputs to 0.17–0.58 in the canonical rows. Outputs are normal-

Table 6: Archetype definitions. Annual IT energy is the weight used in every IT-energy-weighted mean reported in Section 4. ϕ is the flexible-load fraction (Appendix A.3) and α^{base} the base cooling-overhead ratio before the efficiency retrofit (Appendix A.1). District-heating acceptance capacity is driven by the shared uncertainty state's heat demand (Table 7), not by archetype technology, so the reported figure is the same state-driven quantity for every archetype (mean 19.3 MW_{th}; range 10.0–33.9 MW_{th} across the ten states); each archetype's own heat-exchanger capacity and recoverable-heat fraction then determine how much of that shared capacity it can actually use.

Archetype	IT capacity (MW)	Annual IT energy (GWh/year)	α^{base} (–)	ϕ (–)	Recoverable-heat fraction (–)	DH acceptance capacity (MW_{th})
AI/HPC	40.0	345.0	0.12	0.24	0.50	19.3
Hyperscale cloud	60.0	515.1	0.18	0.16	0.45	19.3
Colocation	25.0	189.5	0.30	0.10	0.32	19.3
Enterprise/edge	8.0	61.5	0.42	0.05	0.20	19.3

Table 7: Shared uncertainty states, absolute input levels. El. price and heat price are EUR/MWh(th); flex price is EUR/MW-h; ETS and carbon-damage prices are EUR/tCO2; delay is months.

State	Loc.	El. price	El. vol.	Heat price	Heat dem.	Flex price	ETS	Damage	Delay
1 Central-base	Netherlands	65	18	58	50	18	65	120	6
2 Clean-cool	Norway	48	13	48	62	16	45	80	0
3 High-ETS	Germany	72	24	62	48	21	105	160	6
4 Heat-rich	Denmark	58	20	78	78	20	75	140	0
5 Flex-rich	UK	82	30	52	30	31	80	130	12
6 Cooling-stress	France	70	22	55	38	18	70	110	6
7 Congestion-stress	Belgium	76	27	60	44	24	85	180	6
8 Storage-limited	Sweden	54	19	50	55	17	60	100	0
9 Cost-stress	Netherlands	74	25	56	42	19	90	200	12
10 Service-upside	Denmark	60	21	82	70	29	75	150	6

ized by IT energy or IT capacity where appropriate, and every IT-weighted mean reported below uses the annual IT energies in Table 6 as weights.

Table 7 and Table 8 report the ten shared uncertainty states in full; the scenario generator and seed are released with the code (Code availability, below).

4. Results and Discussion

The PUE–siPUE split is reported first, then qualified through decomposition, compact robustness checks, sensitivity, and face-validity comparison [38, 39].

4.1. PUE and siPUE move differently

Across the canonical study, the two metrics point in opposite directions on the same designs in most rows. Every service-active row raises PUE: all

Table 8: Shared uncertainty states, multipliers applied to archetype base values. Cong. = congestion, Cool. OH = cooling overhead, Free cool. = free-cooling availability, Heat rec. = heat-recovery fraction, HP COP = heat-pump coefficient of performance, Flex avail. = flexibility availability, Storage = storage availability, Capex and Land are cost multipliers.

State	Cong.	Cool. OH	Free cool.	Heat rec.	HP COP	Flex avail.	Storage	Capex	Land
1 Central-base	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
2 Clean-cool	1.20	0.82	1.25	1.10	1.08	0.95	1.10	0.90	0.80
3 High-ETS	0.90	1.05	0.95	0.95	0.95	0.90	0.90	1.00	1.10
4 Heat-rich	0.85	0.92	1.15	1.15	1.05	1.00	1.00	1.05	0.95
5 Flex-rich	0.75	1.08	0.90	0.90	0.90	1.20	1.00	1.00	1.05
6 Cooling-stress	1.05	1.28	0.70	0.85	0.92	0.90	0.90	1.10	1.00
7 Congestion-stress	0.60	1.12	0.90	1.00	0.95	1.05	0.90	1.05	1.10
8 Storage-limited	1.10	0.88	1.20	1.05	1.05	0.95	0.35	1.00	0.90
9 Cost-stress	0.80	1.15	0.85	0.90	0.90	0.90	0.80	1.35	1.30
10 Service-upside	0.95	0.95	1.10	1.15	1.08	1.15	1.15	0.90	0.95

40 baseline rows and all 40 robust rows show higher PUE than the services-disabled comparator. siPUE moves the other way in 36 of the 40 baseline rows and in 23 of the 40 robust rows, where the split is closer to even. Figures 6 and 7 show this by archetype. PUE rises with integration intensity in every case, while siPUE stays flat or falls for the same designs.

Table 9 reports the canonical averages behind this split, weighted by each archetype's annual IT energy (Table 6). The *facility-energy burden* of a design is its PUE increase over the comparator, $\Delta\text{PUE} = \text{PUE}^d - \text{PUE}^0$, and the *credit* is the PUE-to-siPUE gap, $E_{\text{credit}}/E_{\text{IT}} = \text{PUE}^d - \text{siPUE}^d$; their ratio measures how much energy-equivalent system value each additional unit of facility electricity buys.

The services-disabled comparator, with the same efficiency retrofit and carbon-cap treatment as the service-active cases, gives PUE and siPUE values of 1.17 and is the reference case. Against it, the baseline design raises average PUE to 1.24 while lowering average siPUE to 1.14, earning 5.60 EUR/MWh IT of delay-adjusted system benefit, about 10% of the reference electricity price and a credit about 1.40 times its facility-energy burden. The robust design raises PUE further, to 1.34, and lowers siPUE to 1.15, earning a larger absolute benefit, 10.56 EUR/MWh IT, about 19% of the reference price, with a smaller credit-to-burden ratio of about 1.10. Baseline therefore has the marginally lower average siPUE, while robust creates more absolute system value at a lower benefit-to-burden ratio. This pattern is consistent with diminishing marginal system value as integration intensity scales and with the non-monotonicity property noted in Section 3.

The same movement is unevenly distributed across archetypes, and the ranking changes between the two service-active cases. Hyperscale cloud earns

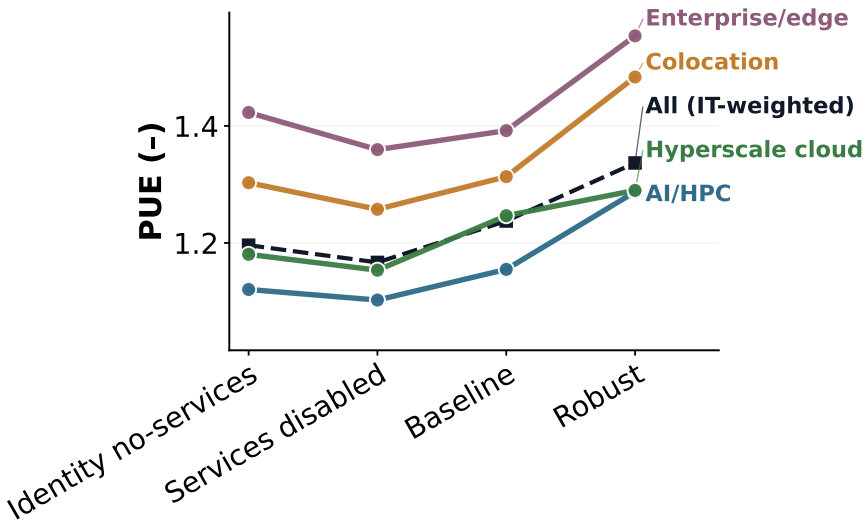


Figure 6: PUE by archetype across the reported design cases and the identity solve, with the IT-weighted mean shown dashed. The identity solve is included to show the metric identity: at that point, PUE here and siPUE in Figure 7 take the same value by construction; it is not a design case (Table 5).

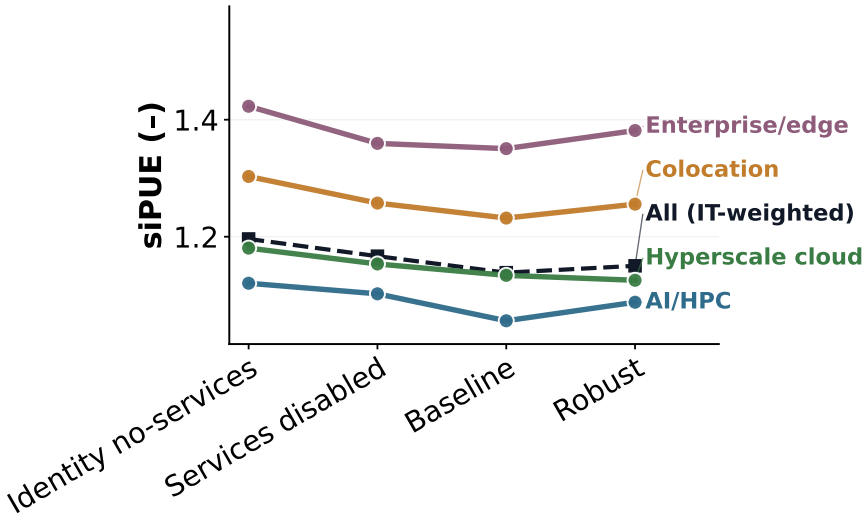


Figure 7: siPUE by archetype across the reported design cases and the identity solve, with the IT-weighted mean shown dashed. The same jointly optimized designs that push PUE up across every archetype (Figure 6) leave siPUE flat or falling here, because siPUE credits the independent system-cost benefit omitted by PUE.

Table 9: Compact metric benchmark across the 40 canonical rows, weighted by each archetype's annual IT energy. Carbon, heat, and emissions are kept as separate ledgers with their own labels. Differences and ratios quoted in the text are computed from unrounded values and then rounded, so they need not equal the difference of the two-decimal entries shown here; for example the unrounded robust burden is +0.18, while the rounded PUE column implies +0.17.

Design case	PUE	Delay-adj. siPUE	Post-constr. siPUE	System benefit (EUR/MWh IT)	Benefit excl. carbon (EUR/MWh IT)
Services-disabled comparator	1.17	1.17	1.17	0.00	0.00
Baseline	1.24	1.14	1.13	5.60	3.43
Robust	1.34	1.15	1.14	10.56	5.10

the largest baseline benefit (6.38 EUR/MWh IT), consistent with its larger baseline heat-exchanger and flexibility capacity (Table 11), yet gains the least in absolute terms under robust (to 9.30 EUR/MWh IT). Colocation and Enterprise/edge start lower (4.61 and 2.35 EUR/MWh IT) but more than double under robust (to 12.91 and 9.76 EUR/MWh IT), overtaking Hyperscale cloud. Archetypes that already capture more benefit at baseline have less room left to gain from the extra capacity the robust design adds.

The average movement is modest because the design criterion is cost-regret robustness; Section 4.5 reports the siPUE-regret alternative. Baseline and robust are reported here because cost-regret robustness is the paper's stated design criterion, and their modest average siPUE movement follows directly from that choice.

The credit is an economic conversion, and its size depends on how carbon damage is valued. Section 4.2 shows the average siPUE improvement reverses once carbon damage is removed from the ledger. Figure 8 shows the paired baseline-to-robust movement in more detail, scenario by scenario, relative to the services-disabled comparator. Each of the 40 arrows moves one archetype-state scenario from the baseline design (dot) to the robust design (arrowhead). Every arrow moves right, because PUE rises in every scenario without exception; what varies is the vertical direction. Arrows ending below the zero line are scenarios where the robust design's larger PUE increase still comes with a lower siPUE than baseline. These are the majority of cases, consistent with the 23-of-40 count reported above. Arrows ending above the line are the counter-examples, concentrated among the Colocation and

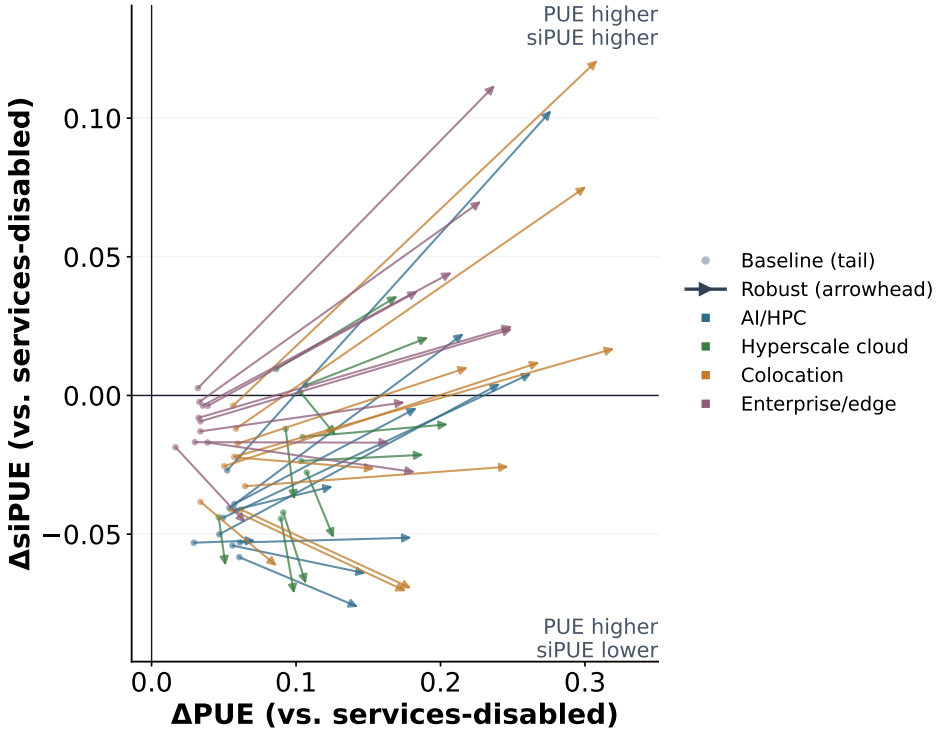


Figure 8: PUE and siPUE tradeoff compass, baseline (dot) to robust (arrowhead), by archetype (color) and scenario, both axes relative to the services-disabled comparator. Quadrant labels on the plot mark the "PUE higher / siPUE lower" versus "PUE higher / siPUE higher" split discussed in the text.

Enterprise/edge scenarios with the longest reach on the PUE axis.

4.2. Credit decomposition and carbon conditionality

Table 10 decomposes the robust siPUE credit. Heat, flexibility, and carbon-damage benefits are partly offset by electricity-resource, peak-capacity, and transmission burdens; construction delay then weakens the net credit. The ablation shows why PUE rises while siPUE can fall. The design uses more facility electricity, but creates system value in the independent ledger.

The carbon-damage line above is also the credit's least stable component. Excluding carbon damage reverses the average finding: removing it from the ledger raises IT-weighted robust siPUE from 1.15 to 1.25, above the services-disabled comparator's 1.17, and only 4 of 40 robust rows remain below the

Table 10: Component ablation of the robust siPUE credit across the 40 canonical rows, weighted by each archetype's annual IT energy. Values are EUR/MWh IT. Components above the delay row are before lifecycle delay; the final row is the delay-adjusted benefit credited by siPUE. Each row is rounded independently from full-precision output, so totals may differ from the displayed component sum by a cent or two.

Component	Contribution	Interpretation
Heat system value	10.58	Avoided independent heat-system cost
Flexibility system value	3.75	Avoided independent flexibility cost
Carbon damage	5.66	Avoided carbon-damage ledger cost
Peak capacity	-1.10	Additional independent peak-capacity burden
Electricity resource cost	-7.56	Additional power-system resource burden
Transmission / network cost	-0.36	Additional network-flow burden
Undelayed system benefit	10.96	Net benefit before delay adjustment
Construction-delay adjustment	-0.40	Lost credit from delayed availability
Delay-adjusted system benefit	10.56	Benefit entering the siPUE numerator

comparator, down from 23 of 40 with carbon damage included. The reported siPUE improvement is therefore conditional on the stated social-carbon valuation, with carbon damage as the component on which the average improvement depends most strongly.

That conditionality concerns how carbon is *valued*. The cap-relaxed resolve tests whether the shared physical carbon cap (Eq. 1) affects how the carbon-damage credit is *earned*. The full two-stage model, including the Stage 1 scenario-perfect benchmarks, the Stage 2 min-max-regret robust design, and the baseline design, was re-solved with the cap constraint removed and the priced carbon-damage term otherwise unchanged.

Baseline quantities are unchanged, while robust quantities move only slightly. The cap-relaxed robust capacities differ from the cap-enforced capacities by at most 0.19 in their reported capacity units. For example, the AI/HPC robust heat-exchanger capacity is 13.52 MW_{th} with the cap and 13.71 MW_{th} in the cap-relaxed run. IT-weighted PUE and siPUE match to three decimal places for baseline (1.24/1.14) and remain 1.34/1.15 for robust. The count of rows below the services-disabled comparator remains 36 of 40 for baseline and 23 of 40 for robust. The cap remains slack. Even with no ceiling, the cost-minimizing baseline and robust designs' own emissions stay below the applied cap in all 40 canonical rows, with the tightest margins equal to 0.003% and 0.015% of the row cap, respectively. The cap-relaxed run attributes the carbon-damage credit in Table 10 to priced carbon-damage valuation, with cap compliance slack in all rows.

4.3. Design portfolios and system value

The baseline case is an assumed incumbent portfolio. A per-MW-IT intensity for heat-exchanger, flexibility, and battery capacity is scaled by an archetype-specific multiplier reflecting typical industry adoption (hyperscale campuses assumed to have invested most in heat recovery and flexibility; enterprise/edge assumed to have invested least), with heat-pump capacity sized to match via the COP-dependent source-to-power ratio and the same locked 15% efficiency retrofit used elsewhere. Table 11 reports both portfolios.

The robust design raises heat-recovery capacity most sharply where the incumbent portfolio was smallest: Colocation moves from 2.50 to 8.41 MW_{th} of exchanger capacity and Enterprise/edge from 0.40 to 2.22 MW_{th}, while Hyperscale cloud, already at 15.00 MW_{th}, stays near that level at 14.60 MW_{th}. Heat-pump and flexibility capacity rise in every archetype. Battery energy moves in the opposite direction for the two largest archetypes, falling from

Table 11: Design capacities by archetype in absolute units: the assumed incumbent (baseline) portfolio, scaled from a per-MW-IT intensity by an archetype-specific adoption multiplier, and the robust min-max-regret design. Both carry the same locked efficiency reduction of 0.15. Round values in the robust battery columns correspond to the capacity bounds in Appendix A.10; a capacity sitting exactly at an imposed bound is determined by that bound before marginal ledger economics can select an interior value.

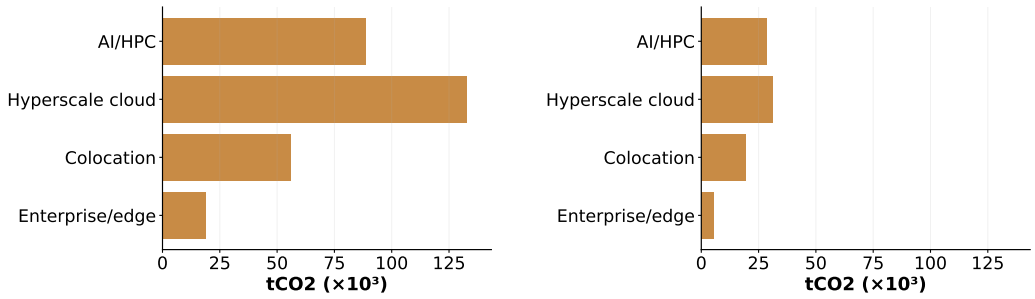
	Baseline (incumbent)					Robust				
Archetype	HX (MW_{th})	HP (MW)	Flex (MW)	Batt. E (MWh)	Batt. P (MW)	HX (MW_{th})	HP (MW)	Flex (MW)	Batt. E (MWh)	Batt. P (MW)
AI/HPC	5.50	2.02	4.80	22.40	5.60	13.52	10.70	6.44	20.00	8.00
Hyperscale cloud	15.00	6.16	4.80	48.00	12.00	14.60	11.56	9.82	30.00	12.00
Colocation	2.50	1.23	1.10	12.00	3.00	8.41	6.66	3.10	12.50	5.00
Enterprise/edge	0.40	0.23	0.19	2.24	0.56	2.22	1.76	0.72	4.00	1.60

22.40 to 20.00 MWh for AI/HPC and from 48.00 to 30.00 MWh for Hyperscale cloud, with battery power held at 0.4 times energy throughout. Relative to the incumbent portfolios, robust portfolios are heat-weighted; the credit decomposition in Table 10 shows heat as the largest single positive component.

Round robust battery capacities indicate binding bounds. Where a capacity sits exactly at an imposed upper or lower bound, the design is pinned by that bound before marginal ledger economics can select an interior value, which also limits how much of the regret behavior in Section 4.5 can be attributed to the uncertainty structure. Appendix A.10 lists the bounds associated with Table 11.

Network congestion is essentially unchanged by these design choices. Averaged over the ten shared uncertainty states, robust-design congestion hours per year range from 2596 (Enterprise/edge) to 2918 (Hyperscale cloud), against 2596–2869 for the services-disabled comparator. The additional facility electricity from heat pumps, flexibility, and batteries does not materially change how often the three-node network is congested. Background load and renewable availability drive congestion here, and the data-center node's own marginal draw is a small share of total line loading.

That ratio also sets the interpretation of the peak and network components. The three nodes represent country groups, so a single facility is a small fraction of nodal load by construction, and C_{peak} and C_{trans} in Eq. 3 are *system-scale* avoided costs evaluated at regional resolution. Their small size in Table 10 follows from the network representation and the regional load scale. Local interconnection constraints require a distribution-level model



(a) Physical power-sector CO2 emissions, robust design, averaged over the ten uncertainty states.

(b) Avoided boiler CO2 emissions from delivered heat, same design and averaging.

Figure 9: Carbon-ledger separation in the robust design. Power-sector emissions and avoided heat emissions are tracked as two distinct physical ledgers, kept separate from the market carbon payment and the independent carbon-damage valuation (Table 3). Both panels share a common horizontal scale, so the flatter profile of avoided heat emissions is directly comparable with the power-sector profile.

and site-specific coefficients. A local application of siPUE would replace both coefficients with distribution-operator values for the specific connection point, where the same facility is a larger share of loading and the peak term can be correspondingly larger.

4.4. Carbon and delay accounting

Carbon-damage valuation can change monetized system benefit without changing physical emissions, so it is retained as a separate ledger item (Table 3) and excluded from the physical sensitivity factors below.

Figure 9 reports the two physical emissions quantities behind that ledger split for the robust design. Power-sector emissions track electricity draw, ranging from 19,088 tCO2 for Enterprise/edge to 132,852 tCO2 for Hyperscale cloud. Avoided heat emissions follow a flatter pattern. AI/HPC and Hyperscale cloud avoid a similar amount (28,793 and 31,144 tCO2) despite Hyperscale cloud's larger power-emissions footprint, because avoided heat scales with installed heat-recovery capacity, while power-sector emissions scale with electricity draw. Neither quantity is the net carbon effect of the design on its own; Table 10 reports the net monetized carbon-damage credit that combines both.

Construction delay affects siPUE only through the availability factor in Eq. 5. It discounts the credit for useful system benefit while leaving the facility-energy accounting intact. The canonical outputs distinguish post-

construction from delay-adjusted siPUE because delay affects the credit and leaves facility-energy accounting unchanged (Section 3). In the robust design, the aggregate loss is about 3.6–3.7% of each archetype's own undelayed credit. Delay enters through the same state-level factor for every archetype; small differences reflect how each archetype's benefit is distributed across the ten states. The absolute loss therefore scales with credit size, from 409 MWh-equivalent/year for Enterprise/edge to 3,150 for Hyperscale cloud. Since a_s enters Eq. 5 linearly, the response to longer delays is close to linear over the asset life. Extending the closed form while holding the robust design and its undelayed benefit fixed, even a decade of delay moves siPUE only about 70% of the way to the degenerate case where $a_s = 0$ and siPUE equals PUE. The tested 0–12 month range is the early, near-linear part of that response.

4.5. Robustness checks

These checks support internal consistency; external validation still requires observed project data. They confirm that siPUE remains finite and nonnegative, reduces to PUE when no services are active, responds monotonically to benefit and delay inputs, and keeps the carbon ledger separate. Held-out and synthetic out-of-sample runs remain feasible. A direct siPUE-regret re-solve also shows that cost-regret robustness differs from siPUE minimization.

In-sample regret is negligible and should be read narrowly. The robust design's normalized regret (Eq. 9) on the canonical scenario set is exactly zero for AI/HPC, Colocation, and Enterprise/edge and of order 10^{-13} EUR/MWh IT for Hyperscale cloud. The worst-case absolute regret is 3.8×10^{-6} EUR over the full lifecycle. One shared design therefore reproduces the scenario-perfect optimum in all ten states it was optimized against. The match limits what the canonical states reveal about regret separation. Within the region of the design space the model explores, the ordering of designs by cost is preserved across the states. The robust design and the ten per-state optima coincide, and the min-max-regret objective adds nothing beyond deterministic optimization in-sample. This result is explained by imposed bounds on several robust capacities (Table 11) and by state perturbations that change prices and availabilities without changing the sign of any technology's marginal value.

The robust design functions here as a well-defined, reproducible design point at which PUE and siPUE can be compared. Validation of min-max regret as a data-center investment rule remains outside this evidence. The

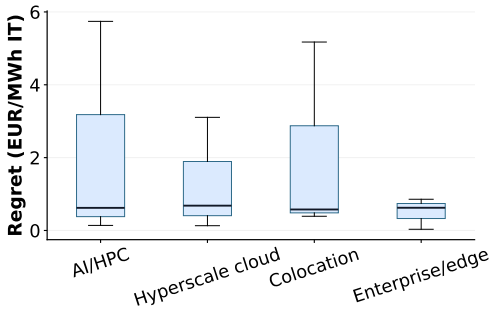
Table 12: Compact internal robustness checks.

Check	Result
Metric properties	Nonnegativity, PUE identity with no active services, delay behavior, benefit monotonicity, scale normalization, and carbon-ledger separation all pass.
Held-out and out-of-sample operation	12/12 held-out rows and 40/40 synthetic out-of-sample rows remain feasible; mean cost regret is 1.51 and 0.29 EUR/MWh IT, respectively.
Direct siPUE-regret alternative	Optimizing siPUE regret directly lowers weighted mean siPUE from 1.15 to 1.06 in-sample and from 1.17 to 1.09 out-of-sample; average lifecycle-cost difference remains below 0.10%.

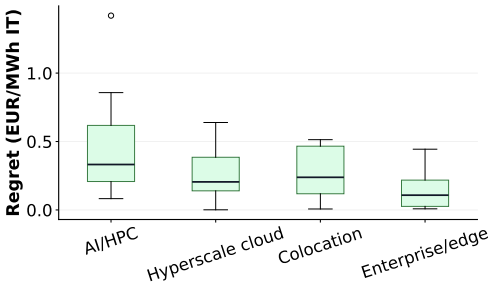
informative robustness evidence is the held-out and out-of-sample regret below, where the design is confronted with states outside its optimization set. A stronger uncertainty test would widen the state space until the in-sample optima separate, which is left to future work.

For the held-out test, the 12 rows use one train/test split per archetype. Scenario-perfect benchmarks and the shared min-max-regret design are resolved on states 1–7 (Table 7), then evaluated on states 8–10 (Storage-limited, Cost-stress, Service-upside) against fresh per-state benchmarks. For the out-of-sample test, each archetype's full canonical robust design is evaluated against ten synthetic scenarios. Price, heat, carbon, congestion, cooling, heat-recovery, flexibility, storage, capex, land-cost, location, and delay inputs are drawn independently over the min–max ranges spanned by the canonical uncertainty tables. Archetype-specific heat-recovery fraction, heat-pump COP, and flexible-load fraction use the same multiplier ranges with the caps and floors in the reference implementation. Fixed per-archetype seeds derived from `cfg.seed` $= 7$ make the 40 rows reproducible from the released code.

Figure 10 shows the distribution behind the two mean regret values above. Held-out regret (panel a) ranges from near zero to 5.74 EUR/MWh IT, with AI/HPC showing both the highest mean (2.17) and the widest spread; out-of-sample regret (panel b) is smaller and narrower overall, with AI/HPC again highest on average (0.47, up to 1.42 EUR/MWh IT). Even the worst individual row (5.74 EUR/MWh IT, held-out AI/HPC) stays below the



(a) Held-out regret. The box marks the median and interquartile range, and the whiskers mark the full range, of cost regret across 12 train/test scenarios, by archetype.



(b) Out-of-sample regret. The box marks the median and interquartile range, and the whiskers mark the full range, of cost regret across 40 synthetic scenarios, by archetype, in the same style as panel (a).

Figure 10: Regret distribution by archetype behind the mean values in Table 12. Panel (a) shows the 12 held-out train/test rows and panel (b) shows the 40 synthetic out-of-sample rows. Both panels show cost regret against each scenario's own scenario-perfect benchmark. Neither panel uses the services-disabled comparator as the reference point.

10.56 EUR/MWh IT system benefit reported for the robust design (Table 9), supporting the mean-regret summary in Table 12.

4.6. Sensitivity analysis

Sensitivity is tested for heat price, flexibility price, cooling overhead, land availability, and construction delay, with the all-archetype result for each level reported in Table 13. Carbon damage remains a separate valuation ledger.

Each sensitivity level re-optimizes the shared design and its dispatch under the perturbed factor, so design and dispatch respond together. In the low heat-price row, system benefit falls from 10.56 to 6.43 EUR/MWh IT and siPUE falls from 1.15 to 1.11. The lower heat price makes heat recovery less attractive, and the re-optimized design shifts toward a smaller, less electricity-intensive portfolio. PUE falls from 1.34 to 1.23, enough to lower siPUE even as the credit shrinks. The row reports both re-optimized design choice and metric response under re-optimization.

Ranked by their effect on robust siPUE, and with Figure 11 sorted the same way, the five factors span about 0.07 for cooling overhead (1.12 to 1.18), about 0.05 for heat price, and about 0.02 for flexibility price and construction delay. Land availability is effectively flat in the all-archetype average. The ordering follows from what each factor perturbs, separate from the width of its input range. Cooling overhead and heat price change the physical

Table 13: All-archetype sensitivity results for the robust design across the 40-row scenario set for each sensitivity case, weighted by each archetype's annual IT energy. siPUE values are delay-adjusted. Small apparent inconsistencies between this table and Figure 11 follow from two-decimal rounding: the tornado is computed from unrounded siPUE values.

Factor	Level	PUE	siPUE	Benefit (EUR/MWh IT)	Land use (%)
Canonical	base	1.34	1.15	10.56	72.3
Heat price	low	1.23	1.11	6.43	56.7
Heat price	high	1.36	1.16	11.46	78.7
Flexibility price	low	1.34	1.17	9.60	70.2
Flexibility price	high	1.34	1.15	10.73	73.6
Cooling overhead	low	1.30	1.12	10.51	71.8
Cooling overhead	high	1.37	1.18	10.65	73.2
Land availability	low	1.31	1.15	9.10	96.1
Land availability	high	1.34	1.15	10.58	60.4
Land availability	no limit	1.34	1.15	10.57	0.7
Construction delay	0 months	1.34	1.14	11.05	73.0
Construction delay	12 months	1.34	1.16	10.04	72.2

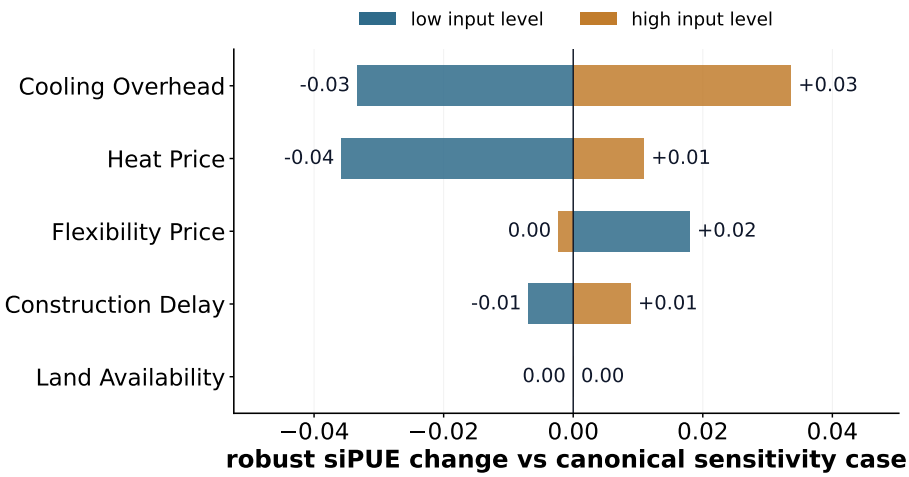


Figure 11: Five-factor sensitivity tornado for robust siPUE, pooled across archetypes as an IT-weighted average per factor. Each factor's low and high input levels are shown as separate bars relative to the canonical value (0), so the direction of each level's effect is explicit. Each level re-optimizes design and dispatch, as discussed above. Carbon damage is excluded from the sensitivity factors.

dispatch the design is built around, so they move PUE and siPUE together. Construction delay only discounts a credit already earned by a fixed design. A 12-month delay against a 15-year lifecycle is a modest fraction of the operating horizon even after discounting, so its effect is smaller and one-directional: it weakens a positive credit.

Land availability behaves differently again, because it moves siPUE only where the limit actually binds. The robust design uses well under the "high" limit, and the unconstrained case mainly changes the denominator used to report land utilization. Only the tighter "low" level forces a land-constrained design, and it binds in 2 robust rows. The same threshold behavior appears across archetypes. Cooling overhead is the most disruptive factor for every archetype, and most of all for Enterprise/edge, whose siPUE swings by 0.150 between the low and high level against 0.040–0.104 for the other three. Land availability, by contrast, produces only small archetype-level movement, largest for AI/HPC and Hyperscale cloud and nearly zero for Enterprise/edge.

4.7. Face validity against published benchmarks

This subsection checks whether modeled PUE and heat-reuse levels are plausible next to published real-project figures, with no model parameter fit to them [38, 39].

The Uptime Institute's 2024 survey reports an industry-average PUE of 1.56–1.58, leading hyperscale sites at 1.2 or lower, and new-build large colocation campuses around 1.4 [38]. The services-disabled comparator gives PUE values of 1.10 for AI/HPC, 1.15 for Hyperscale cloud, 1.26 for Colocation, and 1.36 for Enterprise/edge. The modeled levels are therefore plausible for newer integration-ready facilities and systematically optimistic against the published references, by roughly 0.1 for the colocation and enterprise cases and at the very edge of the reported range for AI/HPC. The direction of that bias matters for how the results should be used: it shifts PUE and siPUE *levels* together, since both share the same numerator and denominator, and leaves the PUE–siPUE *difference* that this paper studies largely unaffected. Absolute levels should not be generalized to the installed fleet.

For heat reuse, the Mäntsälä data center in Finland reuses 75% of its energy as district heat through a dedicated 4.0 MW heat pump [39]. Here, delivered useful heat divided by modeled facility electricity is 0.29–0.31 for AI/HPC and Colocation and 0.21–0.25 for Hyperscale cloud and Enterprise/edge. The published reuse share and the modeled delivered-heat ratio use different possible numerators, denominators, accounting boundaries, and periods, so the published value is quoted as context and kept out of the table. The lower modeled values are consistent with a general-purpose facility that sizes heat recovery alongside flexibility, storage, land, and delay constraints. The Finnish reference is a dedicated heat-export site.

Table 14 summarizes the PUE comparison and bounds how far the results should be extrapolated to specific sites.

4.8. Limitations and future work

The model is a controlled testbed for metric behavior. Absolute NPV values, capacity choices, and KPI levels require project-specific data and calibration before practical use. The quantitative results are bounded by network resolution, in-sample uncertainty separation, objective flatness, and carbon-cap treatment. The three-node network supports consistent regional DC-OPF accounting but omits local interconnection detail, so peak and network credits are system-scale quantities. The ten uncertainty states preserve

Table 14: Face validity of modeled PUE levels against published, real-project figures. The heat-reuse comparison is discussed in the text and kept out of the table because the published reuse share and the modeled delivered-heat ratio use different accounting boundaries.

Quantity	Published reference	Modeled range (this study)
Hyperscale/AI PUE	1.1–1.2 at leading sites [38]	1.10–1.15
Colocation/enterprise PUE	1.4 (new-build) to 1.56–1.58 (industry average) [38]	1.26–1.36

the same in-sample cost ordering across candidate designs. The robust design is therefore best read as a reproducible design point for comparing PUE and siPUE (Section 4.5). The cost objective is nearly flat across materially different designs: re-solving for siPUE regret changes lifecycle cost by less than 0.10%, so small price changes could select a different portfolio. The cap-relaxed re-solve (Section 4.2) leaves baseline unchanged and moves robust capacities only slightly. PUE/siPUE values, comparator counts, and cap slackness remain materially unchanged. The reported carbon credit reflects priced carbon-damage valuation, heat value, and flexibility value.

These limits affect the size of the quantitative credit. The PUE/siPUE distinction remains: PUE can penalize service-active designs because it observes facility electricity without observing the independent system-cost benefit those designs create. Local siPUE applications should replace the independent system-cost unit values with local ones. Archetype-specific evidence remains weakest for hyperscale flexibility, enterprise/edge flexibility, and enterprise/edge heat recovery.

The model treats flexibility as reserve availability; activated scarcity events remain outside the validation scope. Future work should widen the uncertainty state space until in-sample optima separate, add observed transaction data, richer networks, and external validation against measured heat and flexibility transactions.

5. Conclusion

PUE records the services a data center supplies to the surrounding system as facility overhead. Electricity spent on heat pumps, controllable load, and

battery management therefore makes an integrated facility look less efficient even when it creates local-system value.

The accounting rule needed for a PUE numerator credit is developed here. System-Integrated PUE keeps PUE's numerator-over-IT-energy structure and subtracts one delay-adjusted, energy-equivalent credit from an independent system-cost ledger. That ledger combines heat, flexibility, peak-capacity, and carbon-damage value net of the extra electricity, network, and peak cost the design adds. Private revenue, capital cost, and land cost stay outside it. The conversion price is frozen in advance. The credit is zero when no services are active, so siPUE reduces exactly to PUE.

The modeled designs are read differently under siPUE. Every service-active row raises PUE across all 40 canonical scenarios, while siPUE falls below the services-disabled comparator in 36 of 40 baseline scenarios and 23 of 40 robust scenarios. Weighted by annual IT energy, the services-disabled comparator gives a PUE and siPUE of 1.17. A baseline service-active design raises PUE to 1.24 while lowering siPUE to 1.14. A worst-case-cost-regret design raises PUE to 1.34 while lowering siPUE to 1.15 and earning 10.56 EUR/MWh IT of delay-adjusted system benefit, about 19% of the reference electricity price. System value keeps growing as integration intensifies, but more slowly than the electricity needed to deliver it. The more integrated design therefore scores marginally worse. This is the intended behavior: siPUE rewards value-efficient integration.

Removing carbon damage from the ledger removes the average siPUE improvement, so the result depends strongly on the chosen social cost of carbon. The robust design minimizes worst-case cost regret. Minimizing siPUE regret gives a much lower siPUE at a lifecycle-cost increase of less than 0.10%, showing that several different designs have nearly the same cost. SiPUE requires more than facility data. It needs a standardized no-services reference case and externally published avoided-cost and carbon values. PUE should therefore remain the measure of internal facility efficiency, while siPUE should be reported alongside it with the reference price, delay treatment, and reference case clearly stated.

Appendix A. Full operational model formulation

Sections 3 and 4 describe the operational model in prose; this appendix summarizes the hourly linear program that each of the five solve modes of Table 5 is evaluated against. All sums over t run over the representative-day

horizon $t = 1, \ldots, T$ with hourly weight w_t (Section 3); $\Delta t = 1$ h. Parameter values are in Appendix A.10; the reference implementation (Code availability, below) is the authoritative source for the exact constraint set and remaining implementation detail.

A.1 Facility electricity balance and cooling

IT service power follows the archetype's exogenous workload profile, with a symmetric headroom around it when flexibility is active, so IT power can flex up or down by the archetype's flexible-load fraction. Cooling overhead scales with IT power through a cooling-overhead ratio that the discrete efficiency retrofit (0% or 15%, Section 3) reduces; each retrofit level carries its own capex and reduces facility-electricity and opex cost (Section A.9). Facility load at each hour is

$$E_{\mathrm{FAC},t} = \mathrm{IT}_t + \mathrm{COOL}_t + P_t^{\mathrm{HP}} + P_t^{\mathrm{serv}} + \mathrm{CH}_t - \mathrm{DIS}_t, \tag{A.1}$$

the sum of IT power, cooling overhead, heat-pump electricity, and net battery charging, plus a small parasitic overhead for metering and controls that scales with delivered heat, delivered reserve, and installed service capacity. Summing facility load and IT power over the representative-day horizon, weighted by cluster weight, gives the annual E_{FAC} and E_{IT} that define $\mathrm{PUE} = E_{\mathrm{FAC}}/E_{\mathrm{IT}}$.

A.2 Heat recovery and heat-pump operation

When heat recovery is active, recovered heat splits between IT-sourced and cooling-sourced shares in proportion to the cooling-overhead ratio. The flow is capped by the recoverable-heat fraction, installed exchanger capacity, and pipe capacity before entering the heat pump. The heat pump converts that flow into delivered heat and heat-pump electricity at COP- and pipe-loss-dependent ratios, with heat-pump electricity capped by installed heat-pump capacity. Delivered heat can be buffered in a thermal store before sale, following the standard charge, discharge, and state-of-charge relation, bounded by store capacity and cycled shut over the representative-day horizon (zero state of charge at both ends). Delivered heat, net of the store, is sold into up to three district-heating price segments, each capped by a share of district-heating acceptance capacity. When heat is disabled, all heat and thermal-store variables are fixed to zero.

A.3 Flexibility headroom

Flexibility reserve is provided by controllable IT and cooling load and by battery-based reserve. Controllable IT and cooling load is bounded by installed flexibility capacity and by how much headroom remains before hitting the reference IT demand ceiling. Battery-based reserve is bounded by unused inverter capacity and by stored energy above the UPS reserve floor, divided by a minimum reserve-delivery duration (A.4). Total reserve is the sum of both terms; when flexibility is disabled, both are fixed to zero.

A.4 Battery storage and UPS reserve

Battery state of charge follows the standard charge/discharge relation with round-trip efficiency losses, bounded by installed energy capacity, with charge and discharge power each bounded by installed power capacity. State of charge starts at half of installed energy capacity and returns there by the end of the representative-day horizon, closing the cycle once over the horizon as a whole (Section 3). A UPS reserve floor, a fixed fraction of installed energy capacity, holds at every hour; only energy above that floor is available for flexibility reserve.

A.5 Three-node DC-OPF network

The three nodes (with the data center attached to one designated node) balance renewable generation, thermal generation, and DC power flows between node pairs against exogenous background load and, at the data-center node, facility load. Renewable generation is capped by its hourly availability; thermal generation is capped by node-specific capacity. Flows follow a linearized DC power-flow relation through nodal phase angles and are bounded by scenario-specific line capacities. A congestion-utilization variable tracks the most heavily loaded line as a fraction of its capacity and enters the objective with a small coefficient.

A.6 Carbon accounting and cap

Physical emissions combine power-sector emissions from thermal generation with boiler emissions on district-heating demand, net of the boiler

emissions avoided by delivered heat:

$$E^{\mathrm{co2}} = \sum_t \Big[\underbrace{\sum_n \varepsilon_n^{\mathrm{th}} G_{n,t}^{\mathrm{th}}}_{\text{power sector}} + \underbrace{\varepsilon^{\mathrm{boil}} D_t^{\mathrm{dh}}}_{\text{boiler baseline on full DH demand}} - \underbrace{\varepsilon^{\mathrm{boil}} H_t^{\mathrm{sold}}}_{\text{avoided boiler emissions}} \Big] \Delta t \, w_t, \tag{A.2}$$

where the boiler baseline is taken on the full district-heating demand and the avoided term on delivered heat only, so each avoided-heat quantity is credited once. A scenario-specific physical cap enforces $E^{\mathrm{co2}} \leq \mathrm{Cap}_s^{\mathrm{applied}}$ (Eq. 1) in every solve mode except the identity and the unconstrained diagnostic solves. Market carbon cost enters the private lifecycle objective at the scenario's ETS price applied to the same three terms. Carbon damage enters only the independent system-cost ledger at the social cost of carbon (Table 3).

A.7 Land constraints

Land use is the existing facility footprint, plus a term that scales with cooling capacity, plus a fixed per-unit-capacity land allowance for each installed service technology: heat exchanger, heat pump, flexibility, and battery energy and power. When a scenario sets a land limit, land use is constrained not to exceed it; land cost enters the objective as an annualized lease or opportunity cost.

A.8 Peak-capacity cost and construction delay

Peak import is the maximum facility load over the representative-day horizon (Section 3); it enters the private objective as a demand charge and the independent ledger as a peak-capacity value, both linear in peak import. Construction delay does not appear as an operational constraint. It is a lifecycle scalar (Eq. 5) applied to the siPUE credit after the operational problem is solved, and separately to the lifecycle NPV split between pre- and post-commissioning periods.

A.9 Objective

Every design/operations solve minimizes lifecycle market cost, defined as the present-value sum of annual resource, network, carbon-market, and demand-charge costs, net of heat and flexibility revenue and efficiency opex credit, plus capex and annualized land cost. The diagnostic siPUE-regret and

emissions-minimizing solves in Sections 4 and 3 are exceptions that substitute siPUE or physical emissions as the primary objective.

A.10 Parameter values

Table A.15 reports the parameter values used in the canonical run. The reference implementation remains the authoritative source, and the table makes the results reproducible from the manuscript. Archetype-specific quantities are in Table 6, and the ten uncertainty states in Tables 7 and 8.

The 0.01 EUR congestion-utilization weight makes $U_{\text{grid},t}$ equal worst-case line utilization (Appendix A.5). Its scale is negligible relative to lifecycle costs, so dispatch is governed by the economic terms.

The background load and renewable series are scaled from their source resolution to the capacity of this stylized three-node system by a single factor, 0.04, applied identically to load and renewables. The scaling factor is reported because it determines the ratio between facility load and nodal load, and therefore the size of the peak and network terms in Eq. 3.

The capacity upper bounds marked $*$ are absolute fallback ceilings, used only if the per-archetype floor below is disabled. The ceiling actually applied in the canonical run is `battery_capacity_headroom_multiplier` ($=$ 2.0) times the minimum battery energy and power floors, scaled by each archetype's own IT capacity. This gives an effective ceiling of 1.0 MWh and 0.20 MW per MW of IT load. At these settings, robust battery power sits exactly at that 0.20 MW/MW-IT ceiling in all four archetypes (8.00, 12.00, 5.00, and 1.60 MW for AI/HPC, Hyperscale cloud, Colocation, and Enterprise/edge respectively, against IT capacities of 40, 60, 25, and 8 MW). Every reported battery-power value is therefore set by the bound. Battery energy stays at exactly half of its own 1.0 MWh/MW-IT ceiling in every archetype (20.00/40, 30.00/60, 12.50/25, and 4.00/8 MWh). This follows mechanically from the fixed 0.4 power-to-energy ratio combined with power sitting at its bound.

Data and Code availability

The computational implementation supporting this study is available in the BRAVE GitHub repository: https://github.com/sambeets/BRAVE/tree/main/sipue_modelling.

Table A.15: Techno-economic and system parameters, canonical run.

Parameter	Symbol	Value	Unit / source
Lifecycle and financial			
Asset lifetime		15	years
Real discount rate		3.0	%/year
Heat-exchanger capex		220,000	EUR/MW_{th}
Heat-pump capex		600,000	EUR/MW_e
Flexibility capex		200,000	EUR/MW
Battery energy capex		250,000	EUR/MWh
Battery power capex		80,000	EUR/MW
Efficiency-retrofit capex, $\eta = 0.15$		16,800,000	EUR
Land lease / opportunity cost		35.0	EUR/m^2/year
Peak demand charge (private)		50,000	EUR/MW/year
Independent-ledger unit values			
Avoided heat-system cost		55.0	EUR/MWh_{th}
Avoided reserve-capacity value		25.3	EUR/MW-h
Peak-capacity value		50,000	EUR/MW/year
Network / transmission cost		8.0	EUR/MWh
Reference price (canonical)	p_{ref}	56.63	EUR/MWh; identity solve
Social cost of carbon		80–200	EUR/tCO2; by state (Table 7)
Heat side			
Heat-pump reference COP		3.2	—
COP bounds		1.9–3.4	temperature-lift bins 0–120°C
Boiler emission factor	$\varepsilon^{\text{boil}}$	0.222	tCO2/MWh_{th} delivered
Boiler efficiency		0.90	—
Pipe-capacity cap		1.45× state heat demand	MW_{th}; 5% pipe loss
Thermal-store loss and efficiencies	$\lambda^{\text{th}}, \eta^{\text{th}}$	0.001/h, 0.98/0.98	self-discharge; charge/discharge
District-heating price segments	K	3	shares 0.40/0.35/0.25; value mult. 1.20/1.05/0.90
Battery and flexibility			
Round-trip efficiencies	$\eta^{\text{ch}}, \eta^{\text{dis}}$	0.95, 0.95	—
UPS reserve floor	γ^{ups}	0.10	fraction of $K_{\text{BAT,E}}$
Minimum reserve-delivery duration		1.0	h
Network			
Node thermal capacities		5,500	MW, equal at each of N1/N2/N3
Line capacities (calibrated base)		938.3 / 422.6 / 1345.3	MW; $F_{12}/F_{23}/F_{13}$, before the per-scenario congestion multiplier (Table 8)
Background load and renewable series		0.04	scaling factor, load and renewables alike, applied to the source series (Section 4 network-scale caveat)
Congestion-utilization objective weight		0.01	EUR per unit U_{grid}; numerical device only
Design bounds			
Capacity upper bounds*		50 / 25 / 25; 80 / 50	MW_{th}/MW/MW; MWh/MW
Minimum battery energy and power*		0.5 h; 0.10 MW/MW IT	—

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